

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I T. Kaziimulla / 192465046 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Kaziimulla

17
20
R. L. L. /
Total Marks: 30

① The best solution is to use 4G/5G wireless broadband or Satellite internet because high-speed wired internet is unavailable in the rural area. If 5G coverage is available, it should be preferred because it provides better speed and lower latency. Satellite internet can be used as a backup.

Recommended network design:

- Internet Connection: 5G/4G Router as the primary connection
- Backup: satellite or secondary mobile network connection
- Router: Business-grade router with VPN and QoS support
- Switch: Managed Gigabit Ethernet switch for connection office devices
- Wifi: Dual-band wifi 6 access point
- Security: Firewall, VPN, strong passwords, WPA3 encryption and regular updates
- VPN: Establish a secure VPN connection between the branch office and headquarters

Justification:

- Bandwidth: 5G provides sufficient bandwidth for cloud applications and video conferencing.
- Reliability: A backup connection reduces downtime

- Cost: 40% easier to deploy than installing wired infrastructure in remote area.
- Security: Firewall and VPN protect communication with headquarters
- Scalability: Additional access points, switches, and users can be added later

Conclusion: A 5G-based network with a backup connection, firewall, VPN and managed networking devices provides a cost-effective and scalable solution.

② For more than 3000 students the university should use a centrally managed, wifi 6 wireless network with strategically placed access points

- Deploy access points in classrooms, laboratories, libraries and hostels.
- Place Aps after conducting a site survey to avoid unnecessary overlap
- Use the 5GHz band for high-speed connections and lower interference
- Use 2.4GHz mainly where longer coverage is required
- Use different non-overlapping channels for nearby access points

- Use centralised wireless management to monitor and control APs.
- Use WPA3-Enterprise with 802.1X authentication for students and staff.
- Create separate networks/VLANs for students, staff, guests and administrative systems.

Conclusion:

A properly planned wifi 6 network with centralized management channel planning WPA-3 Enterprise, and VLAN Segmentation can provide good coverage and performance while controlling deployment costs.

- ③ The Financial institution should use a segmentation network architecture with firewalls, strong authentication and intrusion detection.

Internet → Firewall → DMZ → Internal Firewall → Core Switch → VLANs.

Network Segmentation:

Create Separate VLANs for:

- Employee Computers
- Customer-sensitive database system
- Online banking/application servers
- Guest users

Security Mechanisms:

- Next-generation Firewall: Controls incoming and outgoing traffic
- IDS/IPS: Detects and blocks suspicious network activity
- MFA: Provides stronger authentication for employee accessing sensitive systems
- VPN: Provides secure remote access
- Access Control Lists: Restrict communication between VLANs.

Balancing the Constraints:

- Security: VLANs, Firewall, IDS/IPS, and MFA protect sensitive information
- Performance: Only necessary traffic is inspected and allowed between network segments.
- Compliance: Strong authentication, logging, access control, and encryption support, security policies
- Budget: Use centralized security monitoring and existing network infrastructure where possible

Conclusion:

A segmented network with VLANs, firewalls, MFA, IDS/IPS, ACLs, and secure monitoring provides strong protection without unnecessarily reducing network performance.